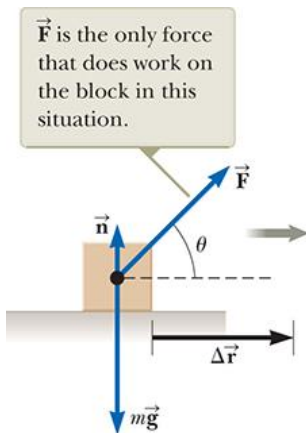


"Long-range goals keep you from being frustrated by short-term failures."

- James Cash Penney (1875 – 1971)



$$\text{Kinetic energy} = \frac{1}{2}mv^2 = \frac{p^2}{2m} \text{ where } p = mv$$

$$\text{Work done: } W = (F \cos \theta) d = \vec{F} \cdot \vec{d}$$

Work-energy theorem:

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \int \vec{F} \cdot d\vec{x}$$

Gravitational PE (constant g) = mgh ,

Gravitational PE (non-constant g over large distances) = $-\frac{GMm}{r}$

Elastic PE = $\frac{1}{2}kx^2$

Serway Fig 7.3 Definition of work done

Applying conservation of energy:

$$\text{K.E.}_i + \text{P.E.}_i = \text{K.E.}_f + \text{P.E.}_f + \text{work done by dissipative forces (if any)}$$

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. determine the work done by a force acting on an object displaced by the force
2. identify situations and when appropriate apply the work-energy theorem to solve problems
3. identify the energies of an object in a system and describe how they are transformed from one type to another
4. apply conservation of energy to solve problems in Physics, including situations where energy is lost as work done against friction or air resistance
5. determine potential energy of conservative forces and vice versa
6. solve problems involving objects moving under the influence of non-constant forces

Work done and kinetic energy (Force and Displacement)

Let us consider what happens when we apply a force F on an object of mass m over a distance d . So here we can conclude that the object will have an acceleration

$$a = F/m$$

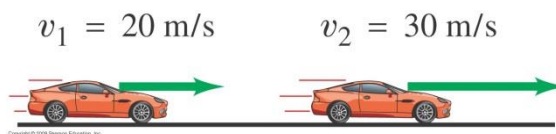
If the object has an initial velocity v_i and final velocity v_f , we can write

$$v_f^2 = v_i^2 + 2ad$$
$$v_f^2 - v_i^2 = 2 \frac{F}{m} d$$

Rearranging, we have

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = Fd$$

The quantity Fd is known as work done and the quantity $\frac{1}{2}mv^2$ is known as (translational) kinetic energy. We use the intuitive way to define energy at the moment – “the ability to do work”. A car (of mass 1200 kg) moving at 20 m/s accelerates to 30 m/s.



What is the kinetic energy of the car when it is travelling at 20 m/s?

What is the change in kinetic energy of the car? What is the work done by the engine to cause this change?

If the distance for this change is 20 m, what is average force exerted by the engine to cause this change in velocity?

Two Analysis of a Baseball Pitch



Giancoli pg 163:

Baseball pitch about to accelerate the baseball to high velocity by applying a force on it

By the time the baseball (mass m) leaves the hand of the pitcher, it would have acquired a velocity v . The gain in momentum of the ball is $p = mv$ and the gain in kinetic energy is $KE = \frac{1}{2}mv^2 (= \frac{p^2}{2m})$.

If we do not have information on the variation of force with time but instead have knowledge of how force varies with distance, we can also analyze the pitching of the baseball, but using kinetic energy and work done W (given by the product of force with displacement along the direction of the force). We will learn about **work-energy theorem corresponding** to these physical quantities

$$W = \int Fdx = KE_{final} - KE_{initial}$$

We will find that this is a very useful principle in the analysis of motions, especially when acceleration is not constant .

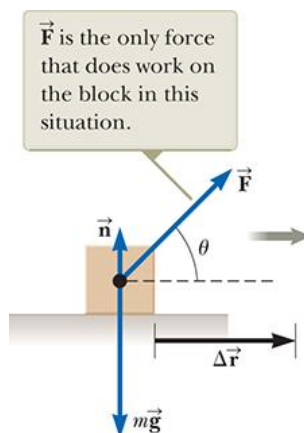
Later, we will learn that the gain in momentum can be analyzed using the impulse-momentum relation

$$J = \int Fdt = p_{final} - p_{initial}$$

The quantity of interest here is impulse which is the product of force and time.

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Work done by a constant force



Serway Fig 7.3 Definition of work done

The work done by a constant force applied is defined as the distance moved multiplied by the component of the force in the direction of displacement:

$$W = (F \cos \theta) d$$

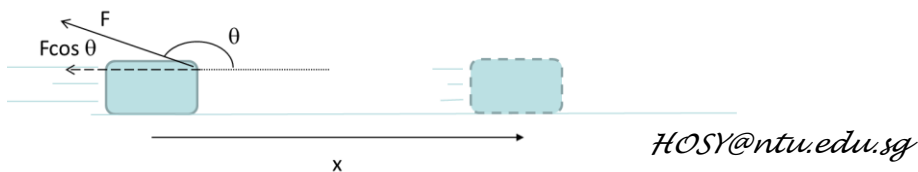
$$= |F| |d| \cos \theta = \vec{F} \cdot \vec{d}$$

Notes:

1. Work done is a scalar and it has units Nm or Joules J .
2. From the definition of work done, we see that when $90^\circ < \theta < 270^\circ$, $\cos \theta$ is negative and W is negative.

We see that when $\theta = 0$, we get the familiar $W = Fd$. We can regard the term $F \cos \theta$ as resolving the force along the direction of displacement.

3. Intuitively, we see

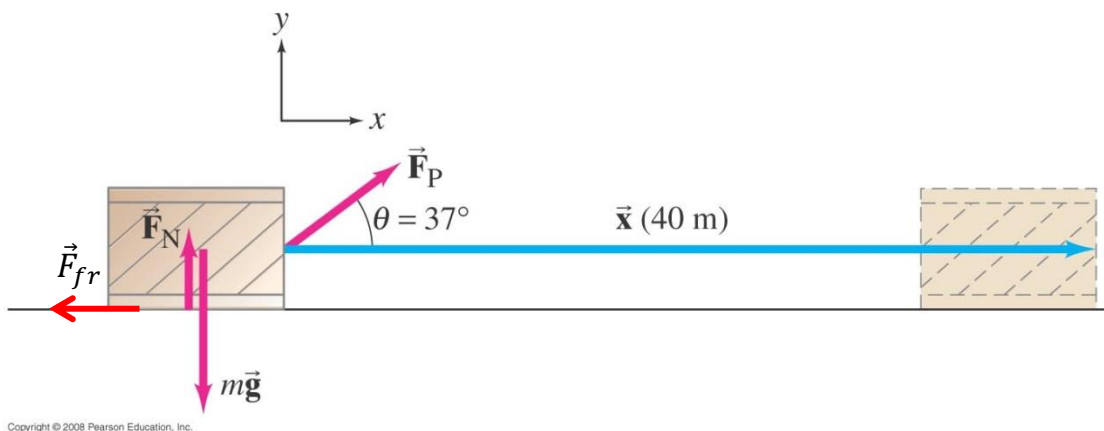


4. When $\theta = 90^\circ$, $W = 0$. So no work is done when the force exerted on the object is perpendicular to its displacement. A classic example is that no work is done by the centripetal force when an object moves in uniform circular motion. This will be discussed later in detail.

$W > 0$	$W < 0$
K.E. of object increases	K.E. of object decreases

Giancoli pg 165 Example 7-1

A person pulls a 50-kg crate 40 m along a horizontal floor by a constant force $F_P = 100\text{ N}$, which acts at a 37° angle as shown. The floor is rough and exerts a frictional force $F_f = 50\text{ N}$. Determine (a) the work done by each force acting on the crate, and (b) the net work done on the crate. (c) What can you say about the change in kinetic energy of the crate at the end of 40m?



$$W_G =$$

$$W_N =$$

$$W_p =$$

$$W_{fr} =$$

$$W_{net} =$$

$$W_{net} =$$

Work done as a scalar product of force and displacement

The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta.$$

Thus, for work done, we can write

$$W = Fd \cos \theta = \vec{F} \cdot \vec{d}$$

Work done as a scalar product of force and displacement

The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta.$$

Thus, for work done, we can write

$$W = Fd \cos \theta = \vec{F} \cdot \vec{d}$$

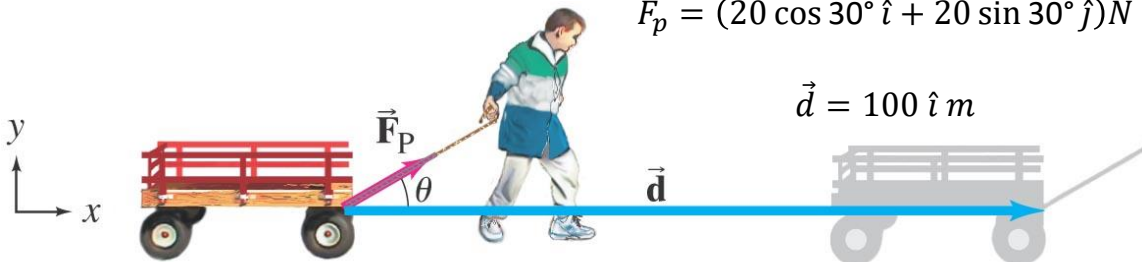
$$\begin{aligned} W &= (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) \cdot (x \hat{i} + y \hat{j} + z \hat{k}) \\ &= F_x x + F_y y + F_z z \end{aligned}$$

since $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$.

Giancoli pg 168 Example 7-4

The force shown has magnitude $F_p = 20$ N and makes an angle of 30° to the ground. Calculate the work done by this force, using the dot product, when the wagon is dragged 100 m along the ground.

$$\vec{F}_p = (20 \cos 30^\circ \hat{i} + 20 \sin 30^\circ \hat{j}) \text{ N}$$



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Example: Airplane taking off

A plane with mass 200000 kg accelerates from rest over a distance of 3200 m on a runway and takes off with a speed of 300 km/h. Estimate the average force acting on the plane over this distance.

Example: Braking

A 800 kg car brake to a stop and left a 25 m long skid mark on the road. If the average braking force is 6000 N, what is the speed of the car when the brakes were applied?

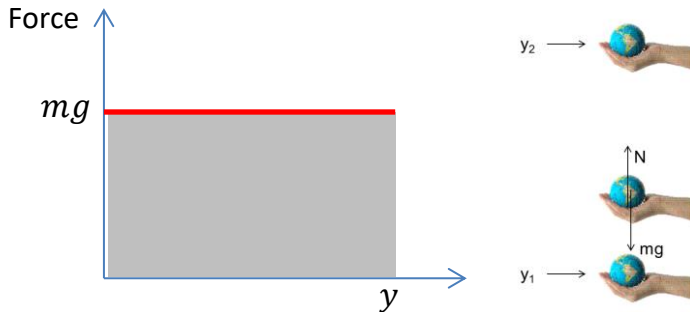


Work Done and Potential Energy

As we will see, in some cases, work done does not always have to result in changes in kinetic energy. It can also be 'stored' as *potential energy*. We will need an "external agent" (a.k.a the hand in our discussion) to do this work. To make the analysis simple, objects are moved at constant velocity in our derivation of the expressions for the various potential energies.

Work Done with constant forces

Moving through small vertical distances (compared to Earth's radius)

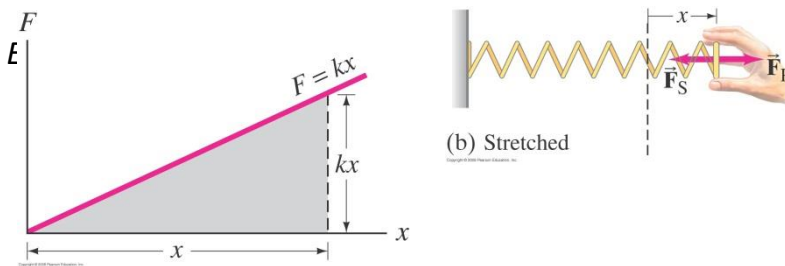


Work done with non constant forces

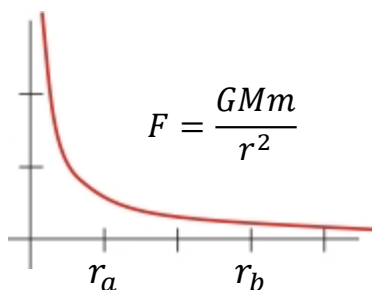
When the force is not constant, the work done is given by

$$W = \int_{x_i}^{x_f} F dx \quad \text{or more generally in more than 1-D, } W = \int_{\vec{x}_i}^{\vec{x}_f} \vec{F} \cdot d\vec{x}.$$

If we have the force-displacement graph, the work done can be obtained by computing the area under the graph. These work done can be stored as potential energy.



Moving through large vertical distances (compared to Earth's radius)



Work done by field and Potential Energy

Consider a falling ball and applying work energy theorem

Taking downwards as -ve

Gain in Kinetic energy = work done by gravity

$$K.E._f - K.E._i = (\text{work done by gravity})$$

$$= -mg \hat{j} \cdot (y_f - y_i) \hat{j}$$

[some algebraic manipulation.]

$$= mg(y_i - y_f)$$

$$= -(P.E._f - P.E._i)$$

We can focus on the relation:

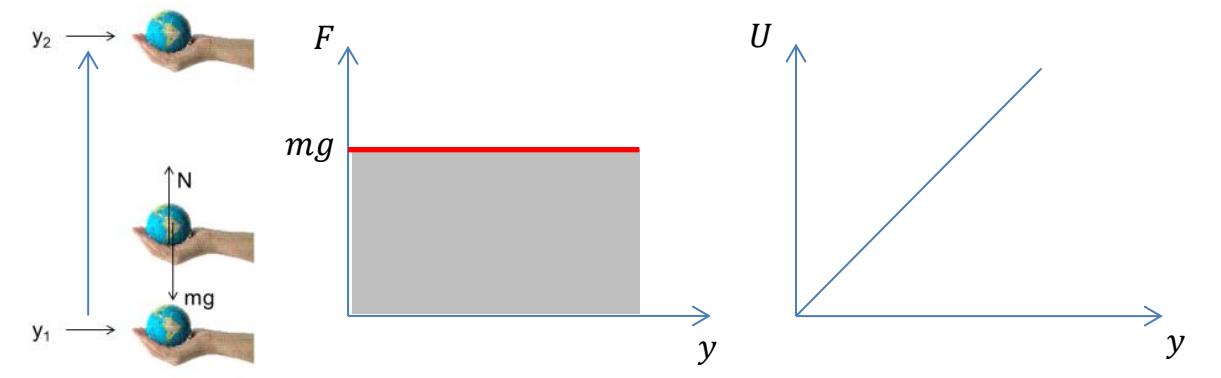
Work done by the field (gravity) = - Change in potential energy

Or

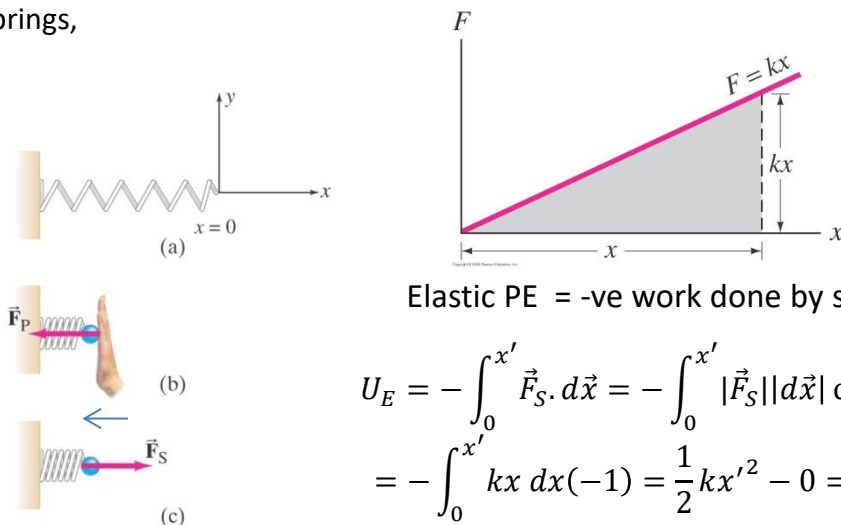
$$\text{Change in potential energy } \Delta PE = - \text{Work done by the field (gravity)} = - \int F_g dy$$

In the illustration below, if a ball is moved up at constant velocity,

the work done by normal force = gain in gravitational potential energy = -work done by gravity



For springs,



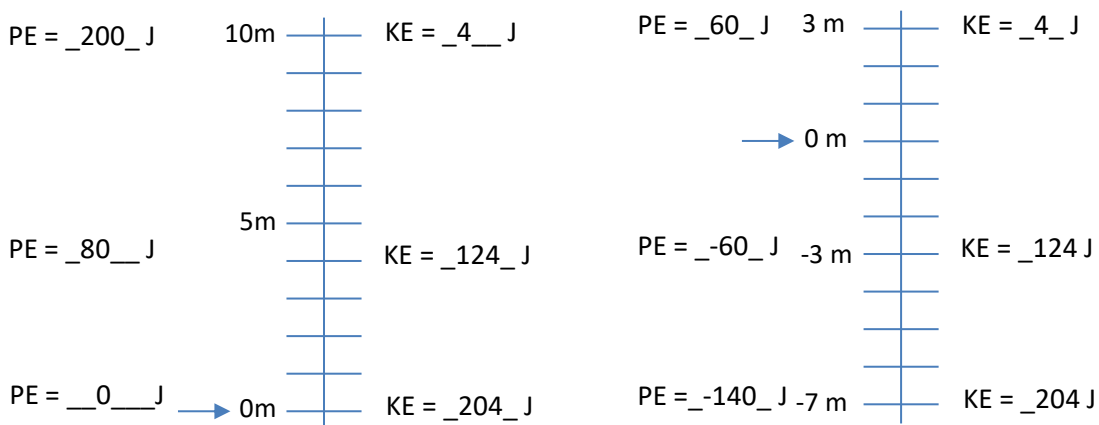
$$\begin{aligned} U_E &= - \int_0^{x'} \vec{F}_s \cdot d\vec{x} = - \int_0^{x'} |\vec{F}_s| |d\vec{x}| \cos 180^\circ \\ &= - \int_0^{x'} kx dx (-1) = \frac{1}{2} kx'^2 - 0 = \frac{1}{2} kx^2 \end{aligned}$$

EXERCISE Giancoli Prob 7.1 (modified) How much work is done by the gravitational force when a 280-kg pile driver falls 2.80 m from rest? What is the kinetic energy of the driver after falling 2.80 m?

EXERCISE Giancoli Prob 8.1 A spring has a spring constant k of 82.0 N/m. How much must this spring be compressed to store 35.0 J of potential energy?

The reference for $GPE=0$ is arbitrary:

As an example, consider throwing a 2-kg mass downwards at 2 m/s from 10 m above the ground. (Take $g = 10 \text{ m/s}^2$ for simplicity.)



We see that even if we choose a different point for zero GPE, we obtain the same kinetic energies at the corresponding levels.

Energy Transfer and Work (Knight)



The athlete does **work** on the shot, giving it kinetic energy, K .

The hand does **work** on the match, giving it thermal energy, E_{th} .



The boy does **work** on the slingshot, giving it elastic potential energy, U_s .

- Energy can be *transferred* between a system and its environment through work and heat.
- **Work** is the mechanical transfer of energy to or from a system by pushing or pulling on it.
- **Heat** is the non-mechanical transfer of energy between a system and the environment due to a temperature difference between the two.

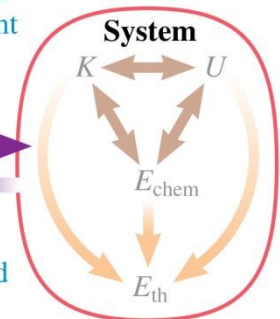
The *environment* is everything that is *not* part of the system.

Environment

Energy is transferred from the environment to the system.

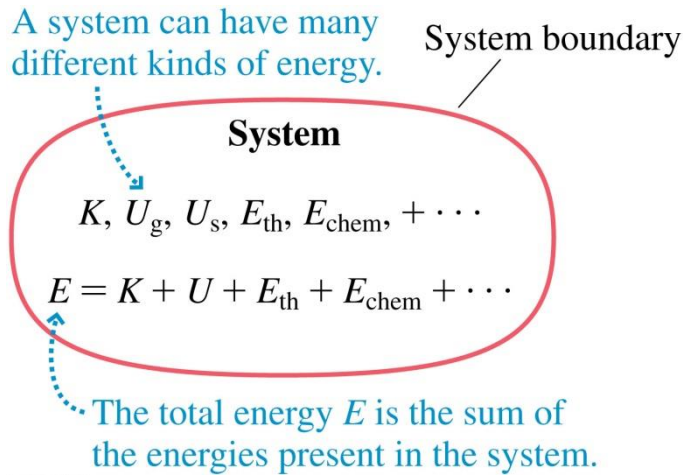
Work,
heat

Energy is transferred from the system to the environment.



Energy (Knight)

Every system in nature has a quantity we call its **total energy** E .



Some important forms of energy are

- *Kinetic energy* K : energy of motion.
- *Gravitational potential energy* U_g : stored energy associated with an object's height above the ground.
- *Elastic or spring potential energy* U_s : energy stored when a spring or other elastic object is stretched.
- *Thermal energy* E_{th} : the sum of the kinetic and potential energies of all the molecules in an object.
- *Chemical energy* E_{chem} : energy stored in the bonds between molecules.
- *Nuclear energy* $E_{nuclear}$: energy stored in the mass of the nucleus of an atom.

Energy (Knight)

Energy of one kind can be *transformed* into energy of another kind within a system.



<https://www.youtube.com/watch?v=xXXF2C-vrQE&t=3s>



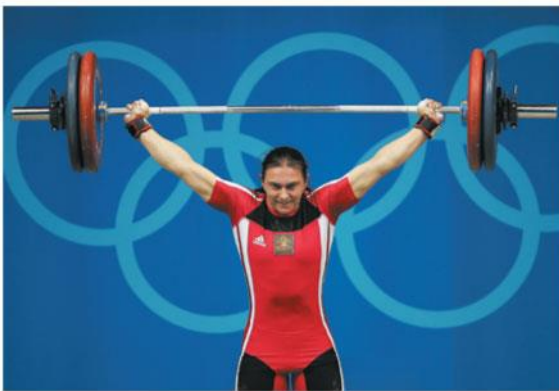
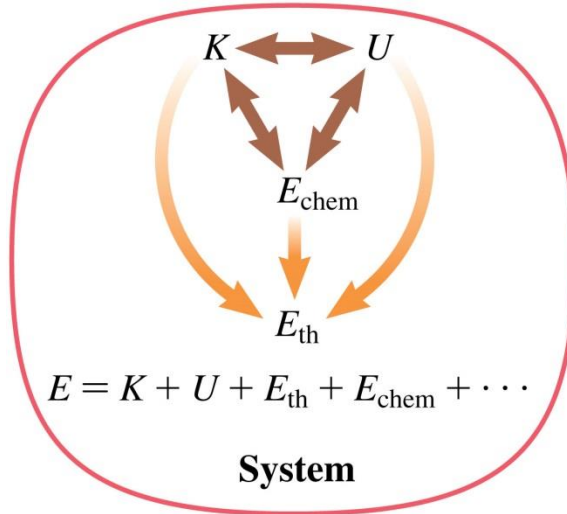
<https://www.youtube.com/watch?v=RLpSluhTb8k&t=2s>



<https://www.youtube.com/watch?v=4r9gmLfpFTg&t=16s>



<https://www.youtube.com/watch?v=hBia7RNARc&t=4s>



The weightlifter converts chemical energy in her body into gravitational potential energy of the barbell.

Elastic potential energy of the springboard is converted into kinetic energy. As the diver rises into the air, this kinetic energy is transformed into gravitational potential energy.



Conservation of Energy

In the example of the falling ball, applying work energy theorem

Change in Kinetic Energy = Work done by the field (gravity)
 = - Change in potential energy

$$K.E._f - K.E._i = -(P.E._f - P.E._i)$$

Note here that the lost in potential energy is equal to the gain in kinetic energy. We can also rearrange the terms and realize that the total mechanical energy at the initial point is equal to the total mechanical energy at the final point:

$$K.E._f + P.E._f = K.E._i + P.E._i$$

This is essentially the conservation of mechanical energy. The kinetic and potential energies (e.g. gravitational, elastic, electric) of the object can transform reversibly into each other. Other types of energy include heat, light and sound.

$$K.E._f - K.E._i = (P.E._i - P.E._f) - \text{other energies dissipated (such as heat, light, sound)}$$

The negative sign for “other energies” indicate -ve work done associated with dissipative forces such as friction, further reducing the gain in KE.

(Consider work done by friction $= \int F_{fric} dx \cos 180^\circ = -\int F_{fric} dx = -\text{energy dissipated}$)

Rearranging, we get

$$K.E._i + P.E._i = K.E._f + P.E._f + \text{other energies dissipated}$$

Nature follows a simple rule:

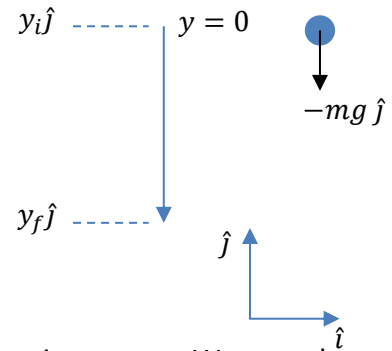
The law of conservation of energy states that the total energy in a closed system is always constant.

In the presence of friction or any other dissipative forces, the above equation can be modified

$$K.E._i + P.E._i = K.E._f + P.E._f + \int F_{fr} dx$$

From the work energy theorem approach, this would be written in the form

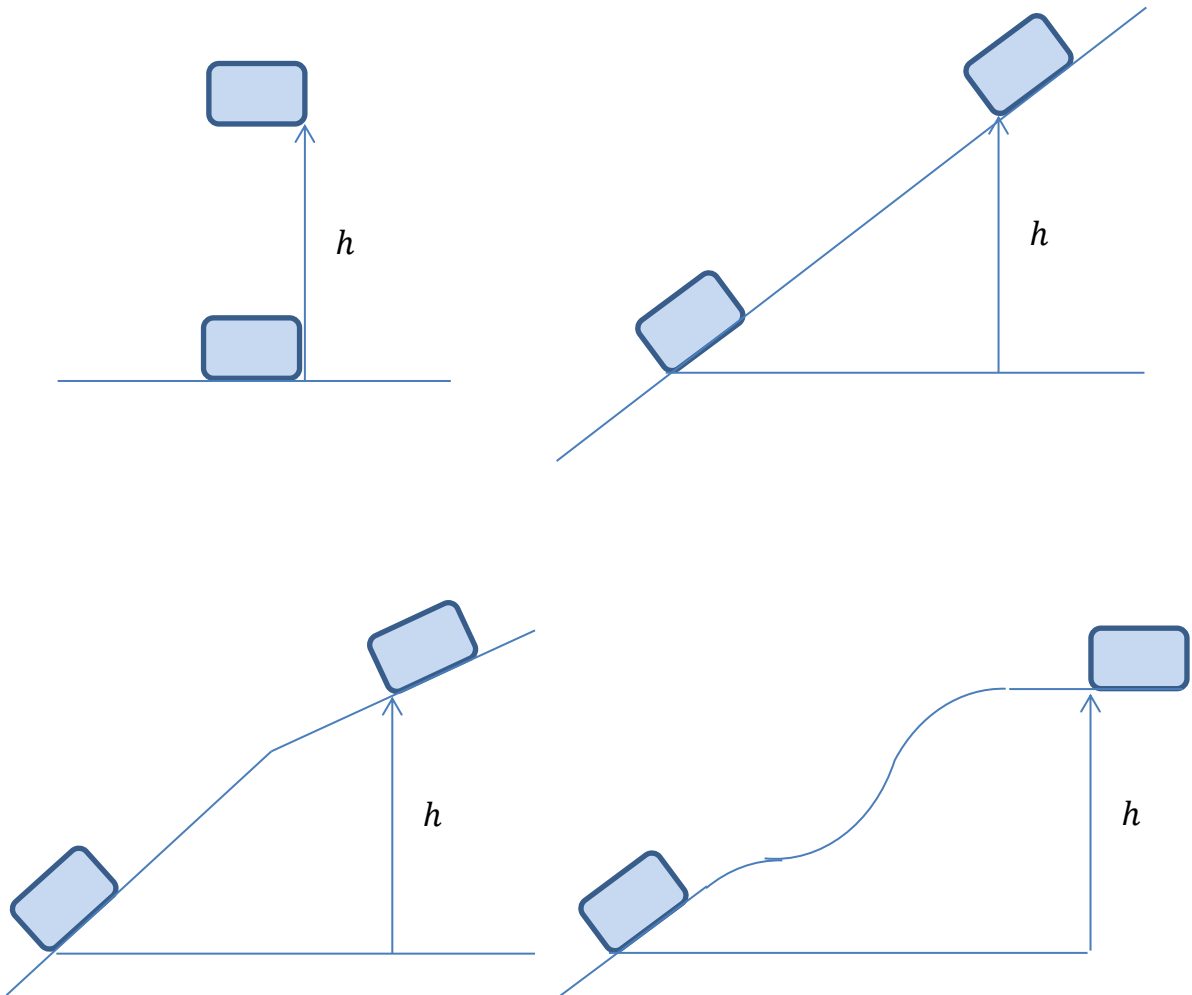
$$K.E._f - K.E._i = \int \vec{F}_{net} \cdot d\vec{x}$$



EXERCISE

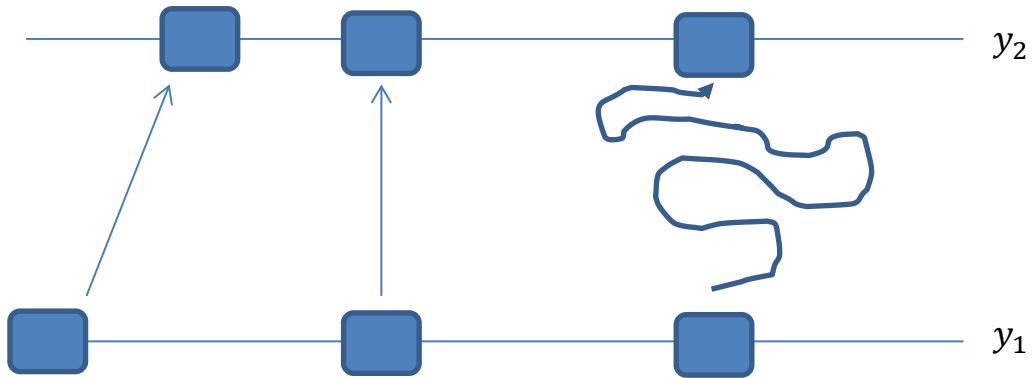
[G8.11] A novice skier, starting from rest, slides down a frictionless 13.0° incline whose vertical height is 125 m. How fast is she going when she reaches the bottom?

Changes in Gravitational Potential Energy and Path Independence



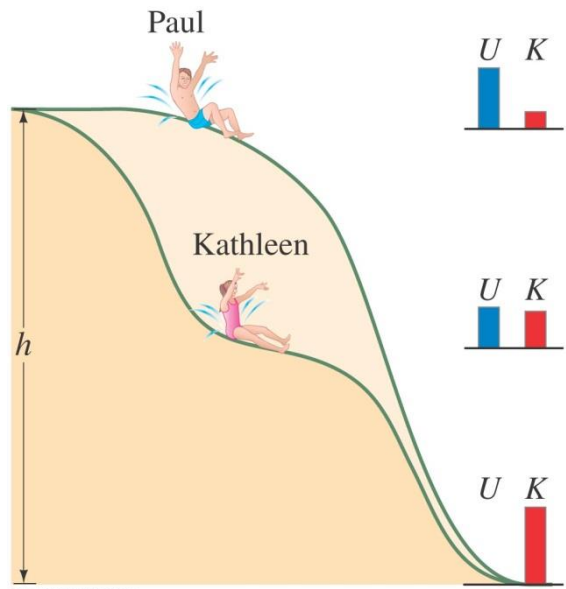
Conservative Force

We note that in the case of gravitational potential energy, the change in energy is path independent and it is only dependent on the starting and end points. In this case, the gravitational force is classified as a conservative force.



Giancoli pg 192 Conceptual Example 8-5

Two water slides at a pool are shaped differently, but start at the same height h . Two riders, Paul and Kathleen, start from rest at the same time on different slides. (a) Which rider, Paul or Kathleen, is traveling faster at the bottom? (b) Which rider makes it to the bottom first? Ignore friction and assume both slides have the same path length.



<https://www.youtube.com/watch?v=VRgxJCWBKk0&t=2s>

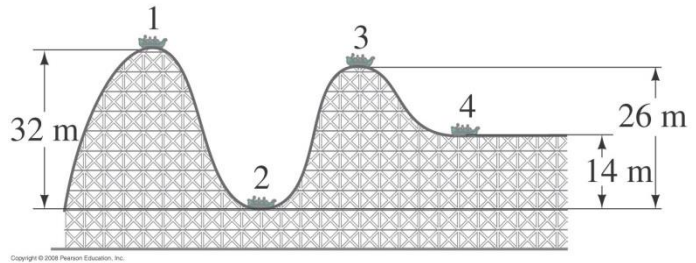


<https://www.youtube.com/watch?v=by2LleVrnyE>

Google homework brachistochrone

EXERCISE

[G8.20] A roller-coaster car shown in the figure is pulled up to point 1 where it is released from rest. Assuming no friction, calculate the speed at points 2, 3, and 4.

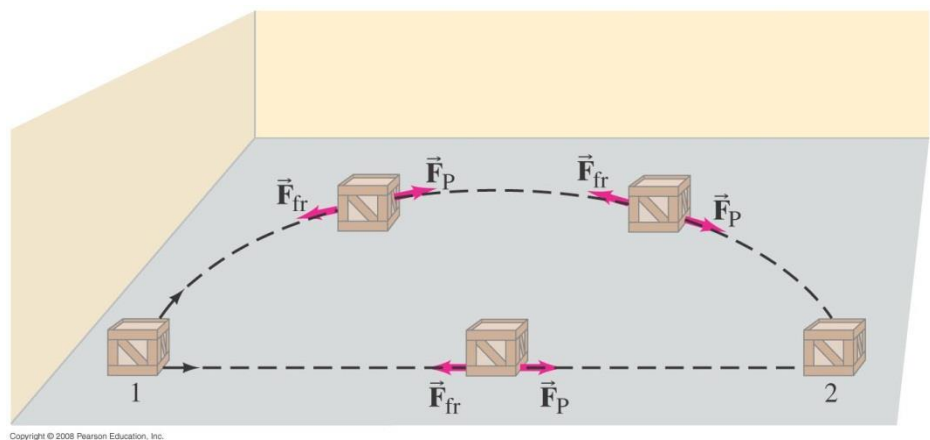


the work done by the force on an object moving from one point to another depends only on the initial and final positions of the object and is independent of the particular path taken.

We also note that the *net work done by a conservative force on an object moving around any closed path is zero*. The work done by a conservative force is recoverable (in particular as kinetic energy).

An example of a non-conservative force is friction.

Examples of non-conservative forces



Giancoli pg 185 Fig 8.3:
 The work done against by a force of constant magnitude F_p is greater round the curved path since $W = F_p d$ and d is larger for the curved path.

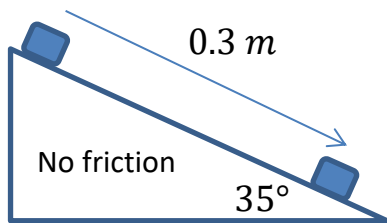
TABLE 8–1 Conservative and Nonconservative Forces	
Conservative Forces	Nonconservative Forces
Gravitational	Friction
Elastic	Air resistance
Electric	Tension in cord
	Motor or rocket propulsion
	Push or pull by a person

Giancoli pg 185 Table 8-1:

Dissipative (Non conservative) Forces: Friction

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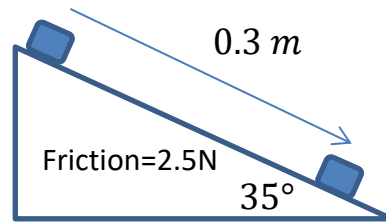
A block of mass 3.0 kg is released from the same height down two different inclined plane with the same tilt of 35° . One is frictionless and the other exerts a constant frictional force of 2.5 N on the block. The displacement of the block along the slope is 0.3 m. Calculate the final velocity in each case.



By conservation of energy,

$$GPE_T + KE_T = GPE_B + KE_B$$

$$mg l \sin 35^\circ + \frac{1}{2}mv_T^2 = mg(0) + \frac{1}{2}mv_B^2$$



By conservation of energy,

$$GPE_T + KE_T = GPE_B + KE_B +$$

$$mg l \sin 35^\circ + \frac{1}{2}mv_T^2 = mg(0) + \frac{1}{2}mv_B^2 +$$

Solving using work energy theorem will yield the same result

$$K.E._f - K.E._i = (P.E._i - P.E._f) - \text{other energies (such as heat, light, sound)}$$

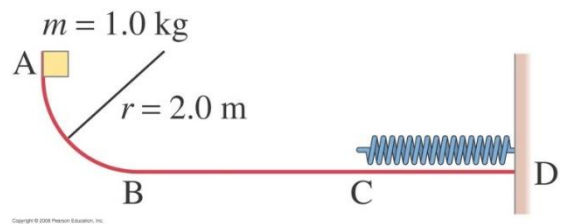
$$K.E._f - K.E._i = (P.E._i - P.E._f) + Fd \cos 180^\circ = (P.E._i - P.E._f) - Fd$$

Class Discussion

[G8.36] Consider the track shown in the figure. The section AB is one quadrant of a circle of radius 2.0 m and is frictionless. B to C is a horizontal span 3.0 m long with a coefficient of kinetic friction $\mu_k = 0.25$. The section CD under the spring is frictionless. A block of mass 1.0 kg is released from rest at A. After sliding on the track, it compresses the spring by 0.20 m.

Determine:

- (a) the velocity of the block at point B;
- (b) the thermal energy produced as the block slides from B to C;
- (c) the velocity of the block at point C;
- (d) the stiffness constant k for the spring.



Power

When a quantity of work ΔW is done during a time interval Δt , the average power P_{avg} , or work per unit time, is defined as

$$P = \frac{\Delta W}{\Delta t}$$

Remarks:

1. Power is also energy per unit time. It is a scalar quantity with units W (Watts) or J/s (Joules per second).

2. Power can also be expressed in a different

$$P = \frac{\Delta W}{\Delta t} = \frac{F\Delta x}{\Delta t} = Fv$$

where F is the force acting on the object when it is moving at velocity v .

For instantaneous power,

$$P = \frac{dW}{dt} = \vec{F} \cdot \frac{d\vec{l}}{dt} = \vec{F} \cdot \vec{v}$$

Example: Vertical Marathon

As part of a charity fund-raising drive, a Singapore marathon runner with mass 50.0 kg runs up the stairs to the top of Marina Bay Sands (194 m) in 8 min. What is her average power output in Watts?

Dissipative (Non conservative) Forces: Scene from The Princess Weiyoung

Prince Tuoba Jun and Princess Weiyoung have to step off a cliff to escape the pursuing assassins. Prince Tuojun pulled out a dagger to scrap against the side of the cliff to break the fall – from the video he appeared to be successful and we assume that after the dagger comes in contact with the side of the cliff there is no appreciable change in kinetic energy of the couple. We estimate the mass of the prince to be 70 kg, princess 50 kg and the height of the cliff to 80 m and effects of air resistance is insignificant. Estimate the temperature change of the 1.20 kg iron dagger (*specific heat capacity* $c = 0.444 \text{ J/g}^\circ \text{C}$) if 70% of the work done by friction becomes thermal energy (the remaining energy became light, sound, blunting the dagger). Take $g = 10 \text{ m/s}^2$ for simplicity.



<https://www.youtube.com/watch?v=7W8kBOMnyQk&t=6s>



<https://www.youtube.com/watch?v=Se6oib28Rg4>

p.s. From the video, the torque of the system does not look too correct.

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Dissipative (Non conservative) Forces: Gravity Light <https://gravitylight.org/>

Problem

Over 1.2 billion people globally have no access to electricity and millions more have an unreliable supply. Instead they use dangerous, polluting and expensive kerosene lamps for light.

How It Works

Combining kinetic and potential energy, GravityLight works by connecting an elevated weight (about 12 kg) — filled with rocks or sand — to a pulley system that slowly powers a generator as the weight falls to the ground. GravityLight is powered by the lift of a weight. As the weight falls it turns a gear train, driving the motor that powers the LEDs. GravityLight doesn't need batteries or sunlight and costs nothing to run. It takes seconds to lift the weight that powers GravityLight, creating 20 minutes of light on its descent.



<https://www.dandad.org/awards/professional/2014/white-pencil/23980/gravitylight/>



<https://www.youtube.com/watch?v=Js c-pQIMxt8&t=3s>

If the mass falls through a height of 2.0 m and assuming that the system is 60% efficient, calculate the power output of the LED light.

"A pessimist sees the difficulty in every opportunity;
an optimist sees the opportunity in every difficulty."
- Sir Winston Churchill (1874-1965)

Outline

Work done: $W = (F \cos \theta) d = \vec{F} \cdot \vec{d}$

Forces in general (incl. non-constant):

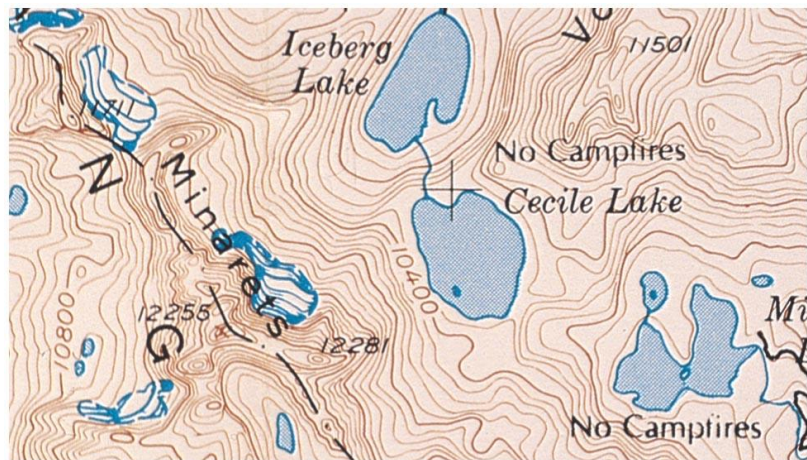
Work done = $\int \vec{F} \cdot d\vec{l}$

Change Potential Energy U due to work done by conservative force $\vec{F}$:

$$\Delta U = - \int \vec{F} \cdot d\vec{x} ; F = - \frac{dU}{dx}$$



<https://www.youtube.com/watch?v=9kNkpQTWpC0&t=3s>



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Giancoli pg 617 Fig 23-19

A topographical map shows continuous contour lines, each of which is at a fixed height above the sea level.

Work done by field and Potential Energy

Earlier, in the example of the falling ball, applying work energy theorem

$$\begin{aligned} K.E._f - K.E._i &= (\text{work done by gravity}) = -mg \hat{j} \cdot (y_f - y_i) \hat{j} \\ &= mg(y_i - y_f) \\ &= -(P.E._f - P.E._i) \end{aligned}$$

We can focus on the relation:

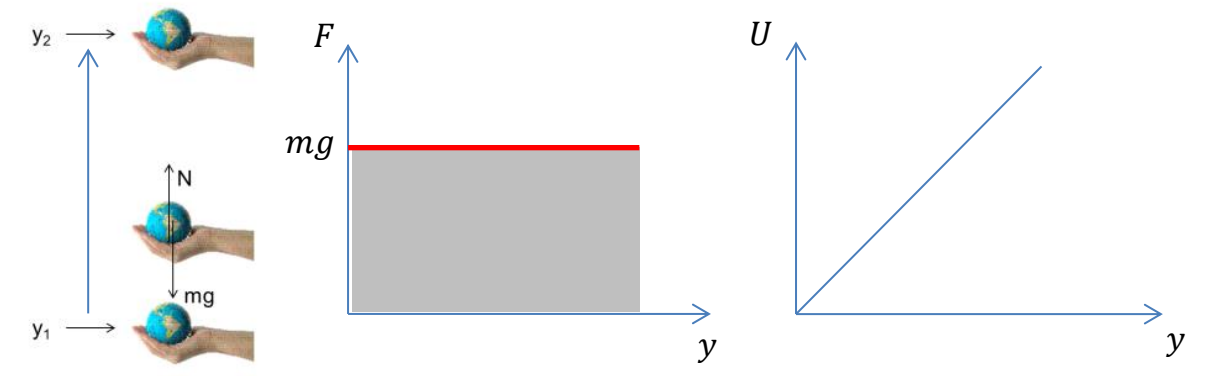
Work done by the field (gravity) = - Change in potential energy

Or

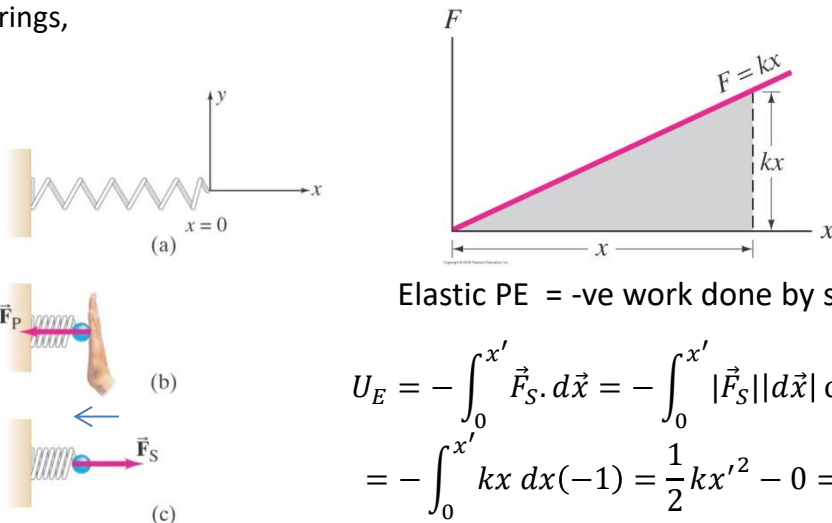
$$\text{Change in potential energy } \Delta PE = - \text{Work done by the field (gravity)} = - \int F_g dy$$

In the illustration below, if a ball is moved up at constant velocity,

the work done by normal force = gain in gravitational potential energy = -work done by gravity



For springs,



Elastic PE = -ve work done by spring

$$\begin{aligned} U_E &= - \int_0^{x'} \vec{F}_S \cdot d\vec{x} = - \int_0^{x'} |\vec{F}_S| |d\vec{x}| \cos 180^\circ \\ &= - \int_0^{x'} kx dx (-1) = \frac{1}{2} kx'^2 - 0 = \frac{1}{2} kx^2 \end{aligned}$$

(Conservative) Potential Energy and Work done

In general, we define the change in potential energy U associated with a particular conservative force $\vec{F}$ as the negative of the work done by that force:

$$\Delta U = U_2 - U_1 = - \int_1^2 \vec{F} \cdot d\vec{l} = -W$$

Remembering that integration and differentiation are inverse operations''

$$\frac{d}{dx} \int F(x) dx = F(x)$$

Thus,

$$F(x) = - \frac{dU(x)}{dx}$$

Giancoli Problem 8.109

A particle of mass m moves under the influence of a potential energy

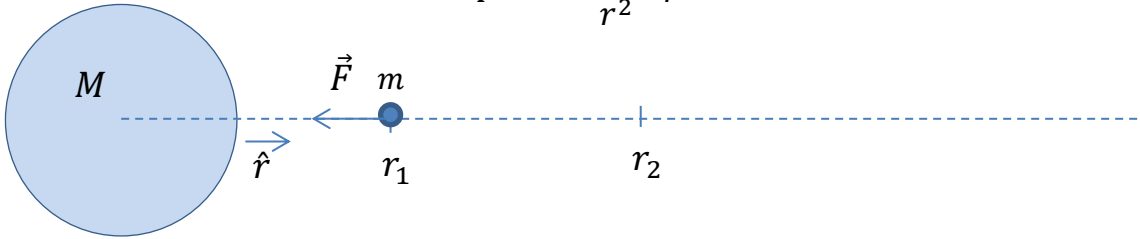
$$U(x) = \frac{a}{x} + bx$$

where a and b are positive constants and the particle is restricted to the region $x > 0$. Find a point of equilibrium for the particle (i.e. net force is zero).

Gravitational Potential Energy and Work done

Consider the vector form of Newton's law of gravitation

$$\vec{F} = -\frac{GMm}{r^2} \hat{r}$$



Consider an object moving from $r_1 \hat{r}$ to $r_2 \hat{r}$. The change in potential energy as given in the earlier discussion is

$$\Delta U = U_2 - U_1 = - \int \vec{F} \cdot d\vec{r} = - \int_{r_1}^{r_2} -\frac{GMm}{r^2} \hat{r} \cdot d\vec{r}$$

Since, the particle with mass m is moving along radial direction (can be generalized to any path in conservative field), $\hat{r} \cdot d\vec{r} = dr$

$$\Delta U = \int_{r_1}^{r_2} \frac{GMm}{r^2} dr = -\frac{GMm}{r} \Big|_{r_1}^{r_2} = -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1} \right)$$

The potential energy at any distance r from M can be written as

$$U(r) = -\frac{GMm}{r} + C$$

where C is a constant. It is usual to choose $C = 0$ so that

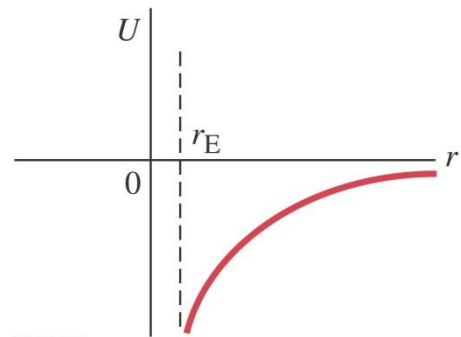
$$U(r) = -\frac{GMm}{r}.$$

With this choice, $U = 0$ when $r = \infty$.

As object approaches M , the potential energy decreases and is always negative.

If r_1 and r_2 near surface of Earth where g is nearly constant, moving from 1 to 2,

$$\Delta U = mg(r_2 - r_1)$$



The total energy of an object mass m moving in a gravitational field (with no other forces present) with velocity v and at distance r from center of mass M is given by $KE + PE = \frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right)$.

Giancoli pg 200 Example 8-12

A box of empty film canisters is allowed to fall from a rocket traveling outward from Earth at a speed of 1800 m/s when 1600 km above the Earth's surface. The package eventually falls to the Earth. Estimate its speed just before impact. Ignore air resistance. (Radius of Earth is 6380 km, mass = 5.98×10^{24} kg).

By conservation of energy,

$$\frac{1}{2}mv_1^2 + \left(-\frac{GM_E m}{r_1}\right) = \frac{1}{2}mv_2^2 + \left(-\frac{GM_E m}{r_2}\right)$$

simplifying,

$$v_2 = \sqrt{v_1^2 - 2GM_E\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} = \dots = 5320 \text{ m/s}$$

Actual velocity is considerably less because of air resistance.

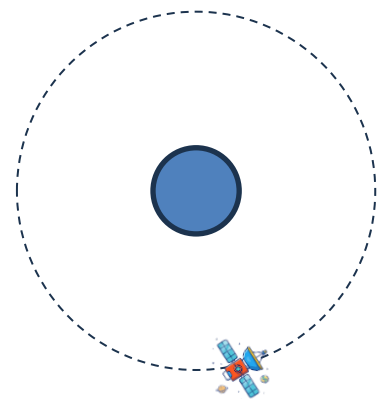
Revisiting Satellites:

Determine the total energy of a satellite of mass $m = 5000$ kg in orbit at a distance $r = 42000$ km.

Gravitation PE:

Kinetic Energy:

Total Energy:



(Conservative) Potential Energy and Work done $\left(F = -\frac{dU}{dr}\right)$

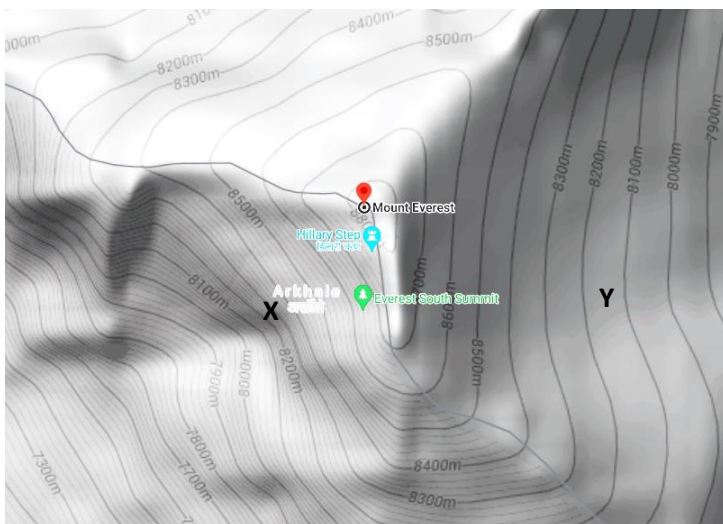
The gravitational potential energy of 2 kg block of masses near the surface of two different planets X and Y are as follows:

30 m		600 J	30 m		1200 J
20 m		400 J	20 m		800 J
10 m		200 J	10 m		400 J
0 m		0 J	0 m		0 J
Planet X			Planet Y		

Which block experiences a larger force? The one on planet X, planet Y or the same? What is the direction of the force?

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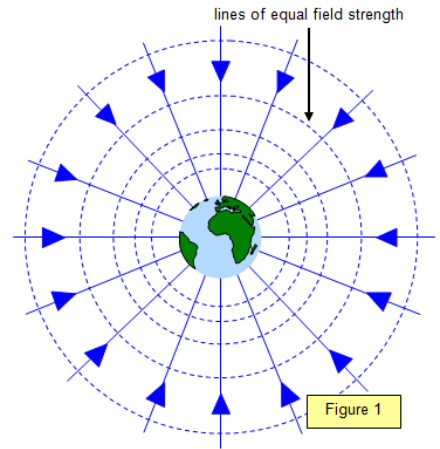
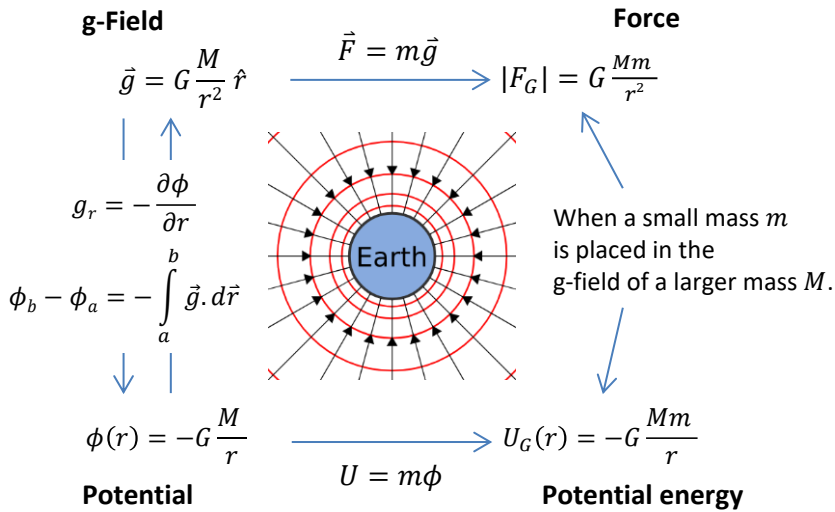
(Conservative) Potential Energy and Work done



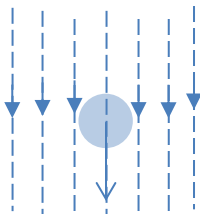
The left is a top view of a terrain map of Mount Everest. If two identical objects are left at point X and Y, which will experience a larger acceleration down the slope? Why?

Summary of Key Ideas

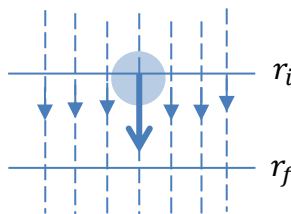
Mass (Radial Fields)



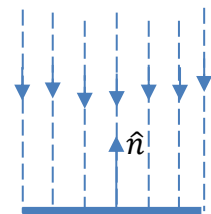
$$\Phi_G = \oint \vec{g} \cdot d\vec{A}$$



Force
& Field



Potential
& Energy



Area A
Total Flux &
Mass / Charge
enclosed

Coming next...

Point Charges (Radial Fields)

